

Enhanced GenPhoto: An I2I Framework for Sequential Camera Control

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Abstract

We introduce *Enhanced GenPhoto*, an Image-to-Image (I2I) framework that resolves scene inconsistency in Text-to-Image (T2I) camera control for iterative and real-world image editing. Unlike prior methods (e.g., *Generative Photography* [3]) that lose subject identity with sequential modifications, our approach integrates *Camera-Conditioned DDIM Inversion* [2] for high-fidelity latent initialization and a *Sequential Multi-Attribute Editing* module for precise, cascaded camera effect application. Our experiments demonstrate superior scene consistency (LPIPS) over SDEdit and T2I baselines, providing robust control for professional virtual photography workflows.

1. Introduction

Recent text-to-image (T2I) diffusion models have demonstrated remarkable generative capabilities. However, in professional photography applications, users require precise and independent control over intrinsic camera attributes, including focal length, aperture (bokeh), shutter speed, and color temperature. A persistent limitation of current T2I models is *scene inconsistency*: modifying a prompt from “50 mm lens” to “wide angle” often leads to drastic changes in subject identity and scene layout, rather than a controlled adjustment of optical perspective.

The recent work *Generative Photography* [3] addresses this issue by introducing a framework for learning disentangled camera intrinsics. Nevertheless, their approach operates primarily in a T2I setting, relying on random noise initialization. As a result, it cannot directly process user-provided real-world photographs, nor can it support iterative and compositional camera edits without degrading scene identity.

In this project, we reproduce the *Generative Photography* framework and substantially extend its capabilities by proposing **Enhanced GenPhoto**, an image-to-image (I2I) framework for sequential camera control. Our method is designed to preserve scene consistency while enabling flexible, multi-attribute camera manipulation. Specifically, En-

hanced GenPhoto consists of two key phases.

Phase 1: Latent Initialization. To enable editing of real images, we introduce a DDIM inversion pathway as a crucial enhancement over the original T2I-based method. This process maps an input image into the diffusion latent space, producing a latent representation z_T and recovering a high-fidelity base content latent z_0 . The resulting latent preserves the scene’s core structure and subject identity, unifying the workflow such that both inverted real images and T2I-generated latents can serve as a consistent foundation for subsequent editing.

Phase 2: Sequential Multi-Attribute Editing. To support compositional camera control, we design a *Parameter Queue* that specifies an ordered sequence of camera edits (e.g., applying bokeh adjustment followed by focal length modification). The base content latent is processed through the GenPhoto editing loop, where the model iteratively applies each camera attribute transformation in sequence. This results in a final image that reflects the cumulative effects of multiple camera setting adjustments while maintaining scene consistency.

To validate our design choices, we conduct extensive ablation studies and compare our inversion-based pipeline against an SDEdit baseline. Experimental results demonstrate that our approach achieves superior preservation of subject identity and scene structure under sequential camera control.

2. Related Work

2.1. Generative Photography

The baseline paper “Generative Photography: Scene-Consistent Camera Control for Realistic Text-to-Image Synthesis” introduces a dataset and training scheme to inject camera physics into diffusion models. It focuses on decoupling semantic content from photographic style using dedicated LoRAs.

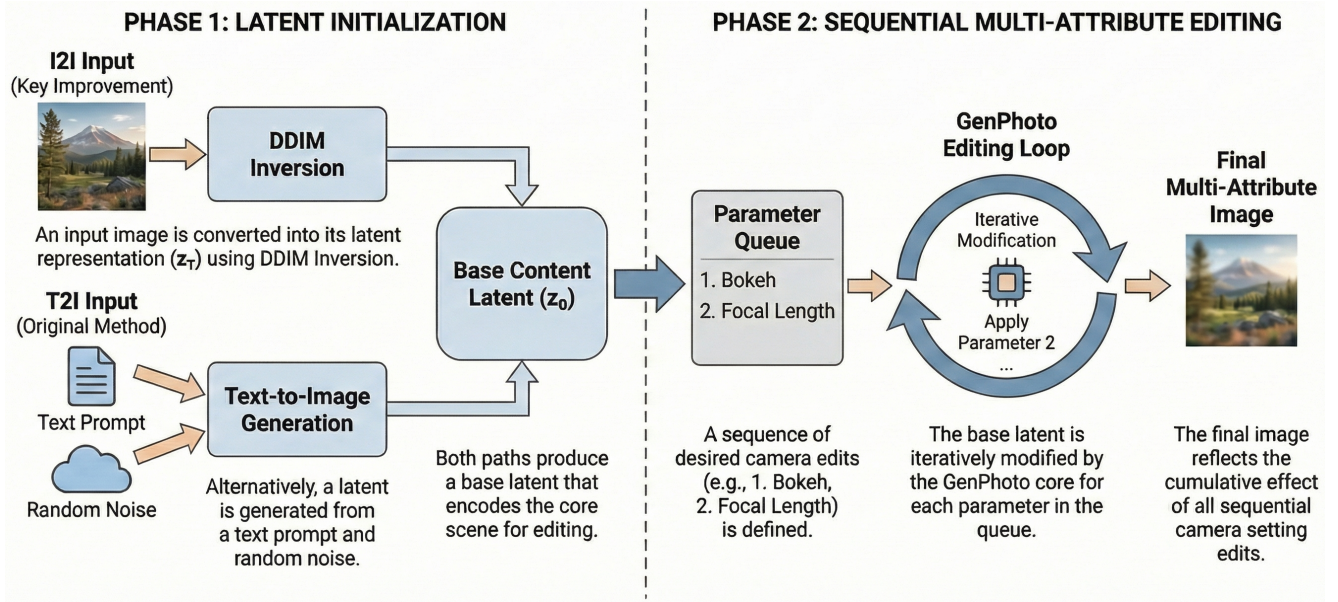


Figure 1. **Overview of the Enhanced GenPhoto Framework.** Our pipeline consists of two phases: **Phase 1 (Latent Initialization)** unifies real-world images (via DDIM Inversion) and synthetic inputs (T2I) into a shared base content latent (z_0). **Phase 2 (Sequential Multi-Attribute Editing)** utilizes a parameter queue to iteratively apply distinct camera effects (e.g., adjusting Bokeh followed by Focal Length), enabling compositional control while maintaining scene consistency.

2.2. Inversion and Consistency

Techniques such as DDIM Inversion and Null-text Inversion are pivotal for editing real or generated images. They reconstruct the initial noise latent that produces a specific image, allowing us to modify the conditioning (e.g., change "f/1.8" to "f/16") while preserving the original composition (SSIM).

2.3. SDEdit

SDEdit [1] serves as a foundational framework for realistic image synthesis and editing without requiring task-specific training. The method operates by first perturbing a given input image with Gaussian noise up to an intermediate timestep t_0 , effectively partially destroying the original signal to strike a balance between faithfulness and editability. Subsequently, it simulates the reverse stochastic differential equation (SDE) process from t_0 guided by a target prompt to recover a realistic image. In our work, we utilize SDEdit as a strong baseline to evaluate the trade-offs between scene consistency and camera control capability compared to our proposed inversion-based pipeline.

3. Method

3.1. Overview

Our proposed framework, Enhanced GenPhoto, addresses the limitations of existing generative photography models by enabling precise, scene-consistent editing on both syn-

thetic and real-world images. Unlike the original GenPhoto [3], which operates strictly in a text-to-image (T2I) paradigm, our method introduces a unified pipeline capable of accepting either random noise or real images as input. The core of our approach is the integration of a Conditioned DDIM Inversion mechanism that preserves the geometric and semantic integrity of the input scene, coupled with a Sequential Multi-Attribute Editing loop for progressive camera control.

3.2. Latent Initialization via Flexible Inputs

The first phase determines the Base Content Latent (z_{start}), which serves as the foundation for subsequent edits. We design a bifurcated input stream where the user explicitly selects the generation mode $M \in \{\text{T2I}, \text{I2I}\}$.

Text-to-Image (T2I) Mode. If the user selects the T2I mode, the process follows the standard latent diffusion initialization. We sample a random Gaussian noise $z_T \sim \mathcal{N}(0, I)$ and use the user-provided text prompt P to generate the initial latent representation via the GenPhoto backbone. This mode is identical to the original implementation [3] and is used when creating scenes from scratch.

Image-to-Image (I2I) Mode with Conditioned Inversion.

For real-world editing, simply adding noise to an input image often results in a loss of fine details and identity shifts. To solve this, we employ a deterministic inversion process

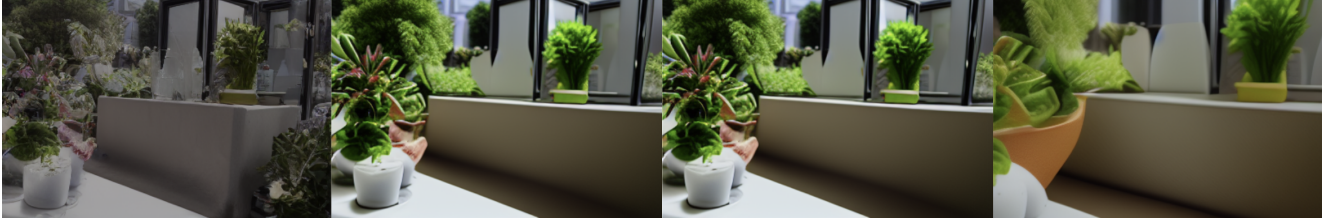


Figure 2. **Qualitative Visualization of Scene Consistency.** Sample output with parameters: Shutter 0.29, Bokeh 5.0, Color 6500.0, Focal 35.0. The image shows a variety of potted plants displayed on a windowsill, with some placed in yellow and white bowls. Despite the application of specific camera attributes, our I2I framework preserves the visually appealing arrangement and the pleasant atmosphere of the original scene, demonstrating superior consistency compared to T2I baselines.

to map a real image I_{src} back to its noisy latent representation z_T . Crucially, standard DDIM inversion is typically conditioned only on text embeddings. However, the visual appearance of I_{src} is heavily influenced by its intrinsic camera parameters (e.g., depth of field). To disentangle these physical attributes from the scene content during inversion, we propose a **Camera-Conditioned DDIM Inversion**. Specifically, we utilize the Differential Camera Encoder introduced in [3]. Let C_{src} denote the camera parameters of the source image. During the reverse diffusion process ($z_0 \rightarrow z_T$), we inject the embedding of C_{src} into the UNet alongside the text prompt. The inversion step at timestep t is formulated as:

$$z_{t+1} = \text{DDIM_Step}(z_t, \epsilon_\theta(z_t, t, P, E_{\text{cam}}(C_{\text{src}}))) \quad (1)$$

where E_{cam} represents the Differential Camera Encoder. By explicitly conditioning on C_{src} , we ensure that the resulting noise z_T encodes the scene content with specific camera effects factored out, allowing for accurate manipulation in the subsequent phase.

3.3. Sequential Multi-Attribute Editing

To enable complex, multi-layered photographic effects, we propose a sequential strategy defined by a Parameter Queue $Q = \{(k_i, v_i)\}_{i=1}^N$, where each tuple represents a target attribute. We formulate this as a cascaded image-to-image translation process. Let I_0 denote the initial image obtained from Phase 1. For each iteration $i \in \{1, \dots, N\}$, the output of the previous step I_{i-1} serves as the new input. We first apply our Conditioned Inversion on I_{i-1} to retrieve the updated latent noise $z_T^{(i)}$. This noise is then processed by the GenPhoto core, conditioned on the new target v_i , to yield I_i . This recursive “invert-and-edit” mechanism ensures that each subsequent camera effect is progressively applied to the most up-to-date visual context, effectively accumulating multiple photographic attributes without content degradation.

3.4. Ablation Study: Inversion Strategy

To justify using DDIM Inversion, we compared it against an SDEdit baseline (Tab. 2).

- **Quantitative vs. Qualitative Reality.** While SDEdit achieves competitive accuracy, DDIM Inversion shows superior structural consistency (lower LPIPS), especially in Bokeh (0.0869 vs. 0.1182). Visual inspection reveals that SDEdit often alters subject identity by adding noise directly. In contrast, our Inversion mathematically retrieves the latent noise, allowing for parameter manipulation while rigorously aligning with the original photograph—essential for professional editing.

4. Experiments

4.1. Implementation Details

- **Datasets and Benchmarks.** We utilize the official validation dataset provided by the Generative Photography framework. The dataset comprises high-quality real-world images and strictly categorized intrinsic attributes: Bokeh, Focal Length, Shutter Speed, and Color Temperature. Standardized parameter lists allow for a fair comparison.
- **Experimental Setup.** Our pipeline is built upon Stable Diffusion v1.5 using pre-trained camera-specific checkpoints. We set the denoising steps to 25 and the Classifier-Free Guidance (CFG) scale to 1.5 to balance fidelity and prompt adherence. All experiments are conducted at 256×384 .
- **Evaluation Metrics.** Following the baseline, we employ three metrics:
 - **Control Accuracy (Acc.):** Pearson correlation between generated attributes and ground truth.
 - **Scene Consistency (Cons.):** Average LPIPS distance between frames to measure identity preservation.
 - **Prompt Following (Follow.):** CLIP similarity score to ensure semantic correctness.

Methods	Bokeh Rendering			Focal Length			Shutter Speed			Color Temperature		
	Acc.	Cons.	Follow.	Acc.	Cons.	Follow.	Acc.	Cons.	Follow.	Acc.	Cons.	Follow.
	Corr↑	LPIPS↓	CLIP↑	Corr↑	LPIPS↓	CLIP↑	Corr↑	LPIPS↓	CLIP↑	Corr↑	LPIPS↓	CLIP↑
Stage 1 (Baseline)	0.9847	0.2559	0.2304	0.9951	0.5019	0.2759	0.4101	0.0680	0.2540	0.6411	0.0688	0.2440
Stage 1 (Ours)	0.8131	0.0702	0.2876	0.9992	0.4513	0.2975	0.6487	0.0893	0.2683	0.7603	0.0767	0.2445
Stage 2	0.7630	0.0745	0.2520	0.6661	0.4503	0.2412	0.6479	0.0537	0.2491	0.7479	0.0606	0.2162
Stage 3	0.5341	0.0892	0.2372	0.6650	0.3968	0.2417	0.5932	0.0693	0.2112	0.7430	0.0709	0.2363
Stage 4	0.6430	0.0623	0.2388	0.6478	0.3298	0.2351	0.7443	0.0748	0.2653	0.7074	0.0887	0.2155

Table 1. **Comparison with the Baseline.** We evaluate our stepwise improvements (Stage 1-4) against the original method (Stage 1 Baseline). Note that our Stage 1 significantly improves LPIPS consistency.

Methods	Bokeh Rendering			Focal Length			Shutter Speed			Color Temperature		
	Acc.	Cons.	Follow.	Acc.	Cons.	Follow.	Acc.	Cons.	Follow.	Acc.	Cons.	Follow.
	Corr↑	LPIPS↓	CLIP↑	Corr↑	LPIPS↓	CLIP↑	Corr↑	LPIPS↓	CLIP↑	Corr↑	LPIPS↓	CLIP↑
SDEdit	0.3859	0.1182	0.2751	0.1806	0.2978	0.2858	0.6441	0.0875	0.2797	0.1625	0.0647	0.2839
DDIM Inversion	0.3863	0.0869	0.2491	0.5534	0.3704	0.2608	0.9416	0.0979	0.2641	0.5745	0.0641	0.2584

Table 2. **Analysis of I2I Editing Methods.** Comparison between SDEdit and our proposed DDIM Inversion approach. While SDEdit achieves higher correlation (Accuracy), DDIM Inversion offers better structural consistency (lower LPIPS) in Bokeh and Color editing.

4.2. Sequential Editing Protocol

To rigorously evaluate our framework without exponential costs, we designed a staged protocol:

- **Stage 1: Initialization Comparison.** We benchmark the initialization quality.
 - *Baseline (T2I)*: Generated from Gaussian noise.
 - *Ours (I2I)*: Inverted from real images via Camera-Conditioned DDIM Inversion.
- **Stage 2–4: Sequential Chain.** To simulate iterative editing (e.g., Shutter → Bokeh), input images for Stage N are randomly sampled from the output of Stage $N - 1$. This evaluates the accumulation of editing effects.

4.3. Quantitative Analysis

Tab. 1 presents the comparison between the baseline and our editing stages.

- **Scene Consistency Improvement.** The most significant advantage is in scene consistency. Stage 1 (Ours) achieves an LPIPS of 0.0702, significantly outperforming the Baseline’s 0.2559. This confirms that conditioned DDIM inversion successfully locks the semantic content, whereas the T2I baseline struggles to maintain identity.
- **Sequential Stability.** In Stages 2–4, LPIPS scores remain stable and low (0.0537–0.0887). This indicates our “invert-and-edit” mechanism prevents structure degradation even after multiple manipulations. CLIP scores also

remain robust (~ 0.24), showing retained semantic understanding.

- **Accuracy vs. Constraint Trade-off.** We note a trade-off in Control Accuracy for geometric attributes like Focal Length (Stage 4 Acc. 0.6478 vs. Baseline 0.9951). The T2I baseline can hallucinate new backgrounds for “wide-angle” prompts, while our method is constrained by the original geometry. However, for style attributes like Color and Shutter, our accuracy remains high (~ 0.70 – 0.76).

5. Conclusion

We introduced Enhanced GenPhoto, an I2I framework that tackles scene inconsistency in iterative and real-world camera control for generative photography. Our two-phase approach, utilizing Camera-Conditioned DDIM Inversion for latent initialization and a Sequential Multi-Attribute Editing module, ensures robust multi-attribute adjustments while preserving scene fidelity. Experiments show superior scene consistency (LPIPS) compared to SDEdit and T2I baselines, paving the way for professional-grade virtual photography.

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